Literature-implied answer: local complementation vs. Lipschitz retraction

arXiv:2107.11339

Status. Literature-implied answer (nonseparable negative; separable case open).

Source question. In the Introduction of Aviles–Martinez-Cervantes–Rueda Zoca, *Local complementation in Banach spaces and its preservation under free constructions*, arXiv:2107.11339, page 2, the authors recall an open problem posed by Kalton: whether local complementation of a subspace $Y \subset X$ is equivalent to the existence of a Lipschitz retraction $r : X \to Y$.

Identification. The broad formulation, with no separability restriction, has a negative answer by combining two standard facts. First, by the principle of local reflexivity, every Banach space E is locally 1-complemented in its bidual E^{**}. Second, Kalton constructed a nonseparable Banach space E which is not a Lipschitz retract of E^{**}; this is recorded explicitly in Quilis–Rueda Zoca, arXiv:2311.13289, page 4, where the authors use it with local reflexivity to produce a Banach-space local-retract example that is not a Lipschitz retract. Therefore $E \subset E^{**}$ is locally complemented but there is no Lipschitz retraction $E^{**} \to E$.

Related reduction and remaining open part. Hajek–Quilis, arXiv:2106.13554, page 4, state the useful reduction: a closed linear subspace Y locally complemented in a Banach space X is a Lipschitz retract of X if and only if Y is a Lipschitz retract of its bidual Y^{**}. The same passage notes that the corresponding separable bidual-retraction problem remains open. Thus the nonseparable broad form of the problem quoted in arXiv:2107.11339 is answered negatively, while the separable Lindenstrauss/Benyamini–Lindenstrauss form remains open.

Provenance. This is an agent-identified implication, not an original proof and not an explicit answer claimed by the source authors. The run already has a separate attempt note for the separable bidual-retraction problem under arXiv:1606.03926.

References.

- A. Aviles, G. Martinez-Cervantes, A. Rueda Zoca, *Local complementation in Banach spaces and its preservation under free constructions*, arXiv:2107.11339.
- P. Hajek, A. Quilis, *Lipschitz retractions and complementation properties of Banach spaces*, arXiv:2106.13554.
- A. Quilis, A. Rueda Zoca, *(Almost isometric) local retracts in metric spaces*, arXiv:2311.13289.
- N. J. Kalton, *Lipschitz and uniform embeddings into ℓ_∞*, Fund. Math. 212 (2011), 53–69.